Semester Project

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Contents

1 SMC EX260-SDN1 DeviceNet **3**
- 1.1 Physical Switch Settings . . . 3
  - 1.1.1 Communication Speed . . . 3
  - 1.1.2 DeviceNet Address . . . 3
- 1.2 Messaging Protocol . . . 3
  - 1.2.1 Connection ID . . . 3
  - 1.2.2 Data Field . . . 4
  - 1.2.3 Setup Flowchart . . . 5
- 1.3 Duplicate MAC ID Check . . . 5
- 1.4 Master/Slave Connection Management . . . 6
- 1.5 Master's Explicit Request . . . 7

2 Keller PAA-4LD I^2C **9**
- 2.1 Request Measurements . . . 9
- 2.2 Read Measurements . . . 9

3 Autonics TM4-N2SB RS485 **9**
- 3.1 Function Codes . . . 9
- 3.2 Register Addresses . . . 10

4 Elemental Analyzer (EA) TCP/IP **10**

5 Graphitization **10**
- 5.1 Conditioning . . . 10
  - 5.1.1 Oxidize Iron . . . 10
  - 5.1.2 Clean Trap . . . 11
  - 5.1.3 Leak Test . . . 11
  - 5.1.4 H2 Cleaning . . . 11
  - 5.1.5 Correct Offset . . . 11
- 5.2 Load Samples . . . 12
  - 5.2.1 Sampling . . . 12
  - 5.2.2 Load Reactor . . . 12
  - 5.2.3 Clean Trap . . . 13

5.3 Reaction . . . 13
5.3.1 Reaction . . . 13
5.3.2 Remove Water . . . 13

6 Graphical User Interface (GUI) 14
6.1 Control . . . 14
6.2 Schematic . . . 14
6.3 Graph . . . 14
6.4 Table . . . 14
6.5 Buttons and Displays . . . 14

7 Threading 14

8 Interfaces 15
8.1 SMC . . . 15
8.2 Keller . . . 15
8.3 TM . . . 16
8.4 Peltier . . . 16
8.5 EA . . . 16

9 Configurations 16
9.1 SMC . . . 16
9.2 Keller . . . 17
9.3 TM . . . 18
9.4 Peltier . . . 18

10 Future Work 18
10.1 Configurations Tab . . . 18
10.2 Parallel Procedures . . . 18
10.3 Storing Data . . . 18
TODO: Citations, GUI, Configurations, Interfaces, Future Work

1 SMC EX260-SDN1 DeviceNet

1.1 Physical Switch Settings

1.1.1 Communication Speed

The communication speed is set with the physical switches No. 3 and 4. The default value is 125kbps.

Bitrate	Switch No.	
kbps	3	4
125	0	0
250	0	1
500	1	0

Table 1: Bitrates corresponding to Switch Settings

1.1.2 DeviceNet Address

The DeviceNet address is set with the physical switches No. 5 to 10. The default address setting is 63.

Address	Switch No.					
-	5	6	7	8	9	10
0	0	0	0	0	0	0
1	0	0	0	0	0	1
2	0	0	0	0	1	0
...				...		
62	1	1	1	1	1	0
63	1	1	1	1	1	1

Table 2: Addresses corresponding to Switch Settings

1.2 Messaging Protocol

A message consists of a connection ID and a data field. Wait $\geq 1s$ for a response when you send a message.

1.2.1 Connection ID

The connection ID counts 11 bits. The structure depends on the message group. Only group 2 is relevant for this project. The connection ID for group 2 contains the group ID (2 bits), the MAC ID (6 bits) and the message ID (3 bits).

	Group ID		MAC ID						Message ID		
Bit No.	10	9	8	7	6	5	4	3	2	1	0

Table 3: Group 2 Connection ID Structure

Message ID	Function
4	Master's Explicit Request
6	Master/Slave Connection Management
7	Duplicate MAC ID Check

Table 4: Relevant Group 2 Message IDs

1.2.2 Data Field

The data field may include some flags. The R flag indicates a request or response. The F flag indicates a complete or fragmented message. The X flag is a transaction ID echoed in a response to match a request.

Flag	0	1
R	Request	Response
F	Complete	Fragmented
X	-	-

Table 5: Data Field Flags

1.2.3 Setup Flowchart

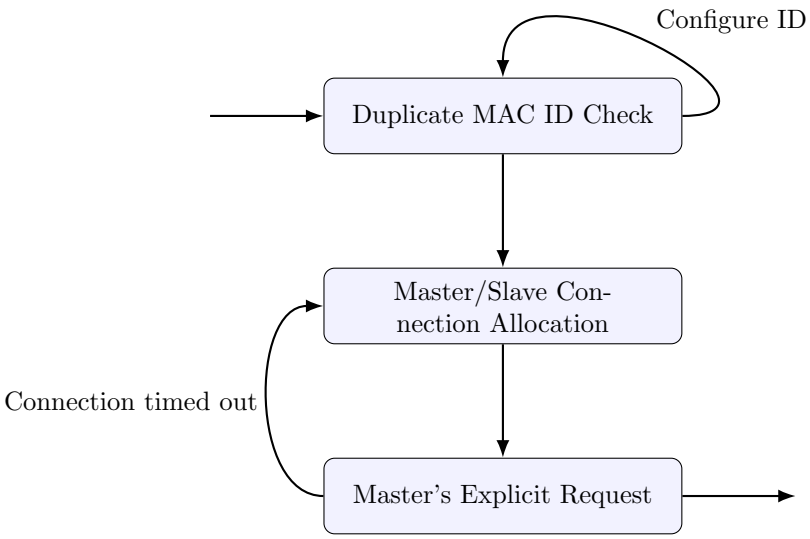


Figure 1: Communication Setup

1.3 Duplicate MAC ID Check

It's inevitable that two modules on the same link will be assigned the same address. It requires human intervention to configure a unique Pi address. The Pi address is written to the Connection ID MAC ID for the duplicate check.

Group ID		Pi Address						Message ID		
1	0	0	0	0	0	0	0	1	1	1

Table 6: Duplicate MAC ID Check Connection ID

Byte No.		
0	R	Physical Port Number
1		Vendor ID
2		
3		Serial Number
4		
5		
6		

Table 7: Duplicate MAC ID Check Data Field

Important: The duplicate MAC ID check has to be transmitted two consecutive times without receiving a subsequent request or response message before establishing a connection.

1.4 Master/Slave Connection Management

Group ID		DeviceNet Address						Message ID		
1	0	1	1	1	1	1	1	1	1	0

Table 8: Master/Slave Connection Management Connection ID

The Pi address is written to the MAC ID in the data field. The service code 0x4B indicates the allocation of a master/slave connection and 0x4C the release. The Class ID 3 directs the message at the DeviceNet object. There exists exactly one DeviceNet object per physical attachement to the DeviceNet. Therefore, the instance ID is always 1. The allocation choice byte denotes an explicit message and I/O connections from the predefined master/slave connection set. The acknowledge suppression bit has to be set to 0 for the release choice.

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x4B]	
2		Class ID [0x03]	
3		Instance ID [0x01]	
4		Allocation Choice	
5		Allocator's MAC ID	

Table 9: Master/Slave Connection Allocation Data Field

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x4C]	
2	Class ID [0x03]		
3	Instance ID [0x01]		
4	Release Choice		

Table 10: Master/Slave Connection Release Data Field

Bit No.	Connection
7	Reserved
6	Acknowledge Suppression
5	Cyclic
4	Change of State
3	Multicast Polling
2	Bit Strobed
1	Polled
0	Explicit Message

Table 11: Allocation/Release Choice

1.5 Master's Explicit Request

Explicit request messages are used to perform operations such as reading and writing attributes. The Pi address is written to the MAC ID in the data field. The service code 0x10 is used to write to a single attribute and the code 0x0E to read from a single attribute. The class ID 9 represents the discrete output points. The instance ID corresponds to the channel.

Group 2		DeviceNet Address						Message 4		
1	0	1	1	1	1	1	1	0	1	0

Table 12: Master's Explicit Request CID

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x10]	
2	Class ID [0x09]		
3	Instance ID		
4	Attribute ID [0x03]		
5	Value		

Table 13: Master's Explicit Set Single Value Data Field

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x0E]	
2	Class ID [0x09]		
3	Instance ID		
4	Attribute ID [0x03]		

Table 14: Master's Explicit Get Single Value Data Field

The class ID 4 represents the Assembly Object. The instance ID 37 corresponds to the 32 solenoid output points. The 4 value bytes act as a mask to set the values for all 32 channels at once.

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x10]	
2	Class ID [0x04]		
3	Instance ID [0x25]		
4	Attribute ID [0x03]		
5	Value		
6			
7			
8			

Table 15: Master's Explicit Set Multiple Values Data Field

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x0E]	
2	Class ID [0x04]		
3	Instance ID [0x25]		
4	Attribute ID [0x03]		

Table 16: Master's Explicit Get Multiple Values Data Field

2 Keller PAA-4LD I^{2}C

2.1 Request Measurements

The command to request a measurement is 0xAC. Wait $\geq 8ms$ before reading the measurement.

2.2 Read Measurements

The response contains the status and the measurement in 3 bytes. The status byte carries a checksum error indicator at bit 2. The two data bytes hold the most and least significant byte. The two data bytes are stitched together and then interpreted with $(p - 16384) * (p_{\max} - p_{\min})/32768 + p_{\min}$ for the pressure. The absolute pressure ranges from $p_{\min} = 0$ bar to $p_{\max} = 3$ bar.

3 Autonics TM4-N2SB RS485

3.1 Function Codes

Code	Function
1	Read Coil Status
3	Read Holding Registers
4	Read Input Registers
5	Force Single Coil
6	Preset Single Register

Table 17: Relevant Function Codes

3.2 Register Addresses

Code	Address for Channel i	Function
1/5	0x0002 * (i-1)	Read/Write Status
3/6	0x03E8 * (i-1)	Read/Write Set Temperature
4	0x03E8 + 6 * (i-1)	Read Temperature

Table 18: Functions

Important: Run=0, Stop=1

4 Elemental Analyzer (EA) TCP/IP

The EA device communicates over the TCP/IP protocol. The messages are sent to the IP address of the EA device via a dedicated port.

Message	Description
STRT	Start device
SLEEPON	Send device to sleep
?STS	Get device status
SEQON	Start processing
SEQOFF	Halt processing
SEQ	Get process status
?SINX	Get sample index `i`
?NAM `i`	Get sample name
?WGH `i`	Get sample weight
?PCT `i` C	Get sample C%
?PCT `i` N	Get sample N%

Table 19: Messages for the EA Device

5 Graphitization

The graphitization process is split into 3 stages: Conditioning the device, loading the samples and the reaction. Each stage has their own procedures. Some procedures can be executed in parallel because they operate on separate sections of the device. The following steps describe the default graphitization process.

5.1 Conditioning

5.1.1 Oxidize Iron

1. Open the line valves

2. Heat the ovens to 500°C
3. Lift the ovens to the reactors
4. Wait 5mins for the irons to preheat

5.1.2 Clean Trap

5. Open the sampling valve for helium from the EA
6. Wait 1min for the trap to be flushed
7. Close the sampling valve

5.1.3 Leak Test

8. Open the reactor valves and the vacuum valve
9. Wait 3mins for the reactors to be evacuated
10. Close the reactor valves and the vacuum valve
11. Check the pressure difference over 5mins

5.1.4 H2 Cleaning

Repeat H2 Cleaning (12. - 15.) ?? times

12. Open the hydrogen valve
13. Wait until one of the reactors reaches 800mbar
14. Close the hydrogen valve
15. Wait 1min

5.1.5 Correct Offset

16. Open the reactor valves and the vacuum valve
17. Wait 20s for the reactors to be evacuated
18. Store the remaining pressure as offset
19. Close the reactor valves and the vacuum valve

5.2 Load Samples

Repeat Load Samples (20. - 54.) for every active reactor

20. Wait for the EA status "STS READY"
21. Store the sample index, name and weight
22. Open the vortex valve
23. Wait for the trap to reach 10°C
24. Close the vortex valve
25. If the sample name starts with "first" (any capitalization) go to Load Samples (20.)
26. If the sample name starts with "pre" (any capitalization) go to Clean Trap (52.)

5.2.1 Sampling

27. Start the EA
28. Wait for the EA status "RUN"
29. If the sample name starts with "runin" (any capitalization) go to Clean Trap (52.)
30. Start processing the sample
31. Wait for the processing status 10
32. Open the sampling valve
33. Wait for the processing status 15
34. Close the sampling valve
35. Stop processing the sample
36. Get the sample C% and N%

5.2.2 Load Reactor

37. Open the trap valve and the vacuum valve
38. Wait 3s for the trap to evacuate
39. Open the reactor valve
40. Wait 10s for the reactor to evacuate
41. Close the vacuum valve

42. Heat the trap to 420°C
43. Wait 10s for the sample to be loaded to the reactor
44. Close the reactor valve and trap valve
45. Store the pressure as CO2 pressure
46. Open and close the vacuum valve and the reactor valve alternately until the pressure reaches ??
47. Store the pressure as CO2 pressure final
48. Open the reactor valve and the hydrogen valve
49. Wait until the pressure reaches ??
50. Open and close the vacuum valve and the reactor valve alternately until the pressure reaches ??
51. Store the pressure as H2 pressure

5.2.3 Clean Trap

52. Open the sampling valve for helium from the EA
53. Wait 1min for the trap to be flushed
54. Close the sampling valve

5.3 Reaction

5.3.1 Reaction

55. Start the peltier coolers
56. Heat the ovens to 580°C
57. Wait 2mins for the reaction
58. Lower the ovens from the reactors
59. Stop the ovens and the peltier coolers

5.3.2 Remove Water

60. Open the reactor valves and the vacuum valve
61. Wait 7mins for the reactors to be evacuated
62. Close the reactor valves, the line valves the vacuum valve

6 Graphical User Interface (GUI)

6.1 Control

The control class is central to the GUI. It holds several other classes such as the devices, the procedures, the samples table, the pressure history plot as well as the schematic. The communication between these classes is connected by the control class.

6.2 Schematic

6.3 Graph

6.4 Table

6.5 Buttons and Displays

7 Threading

The GUI runs on the main thread. Further threads are introduced to keep the GUI responsive at all times. Every device has its own lock. Only one thread at a time can acquire the lock and communicate to the device. There are dedicated threads for every device that regularly poll new data. The procedures are executed in a separate thread that can request writes to the devices. The procedures emit different signals such as a start, end and prompt to interact with the GUI.

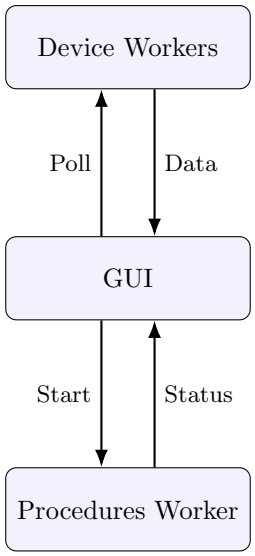


Figure 2: GUI and Worker Interactions

8 Interfaces

8.1 SMC

Signature	Description
`duplicate_check()`	Execute duplicate MAC ID check
`connect()`	Allocate master/slave connection
`disconnect()`	Release the master/slave connection
`get_status(id)`	Get status bit for a valve
`get_status_all()`	Get a list of status bits for all valves
`open(id)`	Open a valve
`open_multiple(idx)`	Open a list of valves
`open_all()`	Open all valves
`close(id)`	Close a valve
`close_multiple(idx)`	Close a list of valves
`close_all()`	Close all valves
`start()`	Perform duplicate check and connect
`stop()`	Close all valves and disconnect

Table 20: Function Signatures and Descriptions

8.2 Keller

Signature	Description
`get_pressure(id)`	Get pressure from a sensor
`get_pressure_all()`	Get pressures from all sensors

Table 21: Function Signatures and Descriptions

8.3 TM

Signature	Description
`get_temperature(id)`	Get temperature for a channel
`get_temperature_multiple(idx)`	Get temperatures for a list of channels
`get_temperature_all()`	Get temperatures for all channels
`set_temperature(id, val)`	Set target temperature for a channel
`set_temperature_multiple(idx, val)`	Set target temperature for a list of channels
`get_set_temperature(id)`	Read target temperature for a channel
`get_status(id)`	Return status bit for a channel
`run(id)`	Start operation for a single channel
`stop(id)`	Stop operation for a single channel
`run_multiple(idx)`	Start operation for a list of channels
`stop_multiple(idx)`	Stop operation for a list of channels

Table 22: Function Signatures and Descriptions

8.4 Peltier

8.5 EA

9 Configurations

9.1 SMC

SMC ID	Pi ID	Bitrate
63	0	125000

Table 23: DeviceNet Configuration

Name	ID
Reactor 1	1
Reactor 2	2
Reactor 3	3
Reactor 4	4
Reactor 5	5
Reactor 6	15
Reactor 7	14
Reactor 8	13
Reactor 9	12
Reactor 10	11
Line 1	6
Line 2	10
Vacuum	7
Trap	8
H2	9
He	16
Vortex	17
-	18
Lift	19

Table 24: Valves Configuration

9.2 Keller

Reactor No.	Address
Reactor 1	0x50
Reactor 2	0x51
Reactor 3	0x52
Reactor 4	0x53
Reactor 5	0x54
Reactor 6	0x55
Reactor 7	0x56
Reactor 8	0x57
Reactor 9	0x58
Reactor 10	0x59

Table 25: Keller Configuration

9.3 TM

Port	Address	Baudrate
/dev/ttySC0	1	9600

Table 26: Instrument Configuration

Name	ID
Trap	1
Oven 1	2
Oven 2	3

Table 27: Channel Configuration

9.4 Peltier

10 Future Work

10.1 Configurations Tab

10.2 Parallel Procedures

10.3 Storing Data